Model MEML 75:

Complete fermion sector, axion-like dark matter, and black hole superradiance
from $T^2/\mathbb{Z}_3$ geometry

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Abstract

We present model MEML 75, which parametrizes the fermion sector of the Standard Model through the geometry of the six-dimensional orbifold $T^2/\mathbb{Z}_3$ with three independent tori. The model has **18 nominal parameters**, of which two are fixed by unphysical overall scales ($a_u = g_1 = 1$) and three (g_3, g_4, g_5) parametrize absolute neutrino masses, which are not constrained by the oscillation data used here. The remaining 13 parameters, together with the ALP misalignment angle θ_i, give **14 effective parameters** that fit **13 fitted observables** (9 quarks + 3 PMNS angles + $\Omega_a h^2$). The model additionally predicts $J_{cp} = 0$, so that it is overdetermined by one constraint: $N_{\rm dof} = (13 + 1) - 14 = 0$ counting the prediction, and $N_{\rm dof}^{\rm fit} = -1$ counting fitted observables only.

The main testable consequence: axion-like dark matter with $f_a = 1.9801 \times 10^{11}$ GeV, $m_a = 28.407$ μeV, $\nu_a = 6.8688$ GHz, and ALP-photon coupling $g_{a\gamma\gamma} = +4.38 \times 10^{-15}$ GeV^{-1} (DFSZ) or -1.13×10^{-14} GeV^{-1} (KSVZ), testable at ADMX/HAYSTAC (0.5–10 GHz). The resulting ALP mass also implies a black hole superradiance window at $M_{\rm BH} \sim 10^{25}$ kg. These are consequences of the chain $\alpha_G \to \varepsilon \to f_a$, not predictions from the orbifold geometry.

Quark sector. A global fit with loss function $\chi^2 = \sum_i ((O_i - O_i^{\rm target})/\sigma_i)^2$ yields $\chi^2 < 10^{-15}$ (numerical precision limit) at $\tau_u = -0.1577944085 + 2.3851782590\,i$, $\tau_d = +0.3693905617 + 1.5410496057\,i$.

Lepton sector. Using a third torus τ_l, the neutrino mass matrix is built from the 3_S contractions of $Y^{(2)}$ and $Y^{(4)}$, plus diagonal contributions from singlet contractions of $Y^{(2)}$, $Y^{(4)}$, and $Y^{(6)}$. The three PMNS angles are reproduced to high precision ($\chi^2_{\rm angles} < 10^{-10}$) at $\tau_l = -0.8534397135 + 1.6846113968\,i$, predicting $\theta_{12} = 33.41°$, $\theta_{13} = 8.54°$, $\theta_{23} = 49.10°$. The model structurally predicts $J_{cp} = 0$; the current NuFIT 5.2 central value $J_{cp} \approx -0.018 \pm 0.005$ is at $\approx 3.6\sigma$, providing a testable prediction for Hyper-Kamiokande and DUNE.

Verdict: the model is *phenomenological*. It interpolates fermion observations with high precision.

1 Introduction

Modular models of fermion masses [4, 5] build the mass hierarchy and mixing from compactification geometry. In model MEML 75, quarks and leptons are localized on three independent tori $T^2/\mathbb{Z}_3$ with moduli τ_u, τ_d, and τ_l. The mass matrices are built from A_4 modular forms of weights $(2, 4, 6)$.

2 Geometry of $T^2/\mathbb{Z}_3$

The orbifold $T^2/\mathbb{Z}_3$ is defined by:

$$z = x + \tau y, \quad ds^2 = dx^2 + dx\,dy + dy^2, \tag{2.1}$$

where $\tau = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$. The Z_3-action $z \to \omega z$ (with $\omega = e^{2\pi i/3}$) has three fixed points, providing a topological explanation for three generations.

3 Bulk coupling α_G (postulate)

$$\alpha_G = \frac{2}{3\pi} = 0.2122065908. \tag{3.1}$$

Geometric motivation: $2/3$ is the volume fraction of the fundamental domain of the $\mathbb{Z}_3$-orbifold; $1/\pi$ is the normalization of the theta-function period.

Geometric motivation via coarse-space topology. The numerical value $2/(3\pi)$ can be read off from the topology of the coarse space of $T^2/\mathbb{Z}_3$. Under the $\mathbb{Z}_3$ action, the torus T^2 collapses onto a sphere: $T^2/\mathbb{Z}_3 \cong S^2$, whose topological Euler characteristic is

$$\chi(S^2) = 2. \tag{3.2}$$

The number of fixed points of the $\mathbb{Z}_3$-action is $N_{\text{fix}} = 3$. Combining these with the theta-function period normalization $1/\pi$ gives

$$\alpha_G = \frac{\chi(S^2)}{N_{\text{fix}}\,\pi} = \frac{2}{3\pi}. \tag{3.3}$$

We emphasize that this is a *geometric motivation*, not a derivation: the relation $\alpha_G = \chi(S^2)/(N_{\text{fix}}\pi)$ is not forced by the orbifold geometry. The status of α_G therefore remains that of a postulate.

4 Suppression factor ε (ansatz)

$$\varepsilon = \left(\frac{\alpha_G}{4\pi}\right)^2 = \frac{1}{36\pi^4} = 2.851662 \times 10^{-4}. \tag{4.1}$$

Two-loop interpretation. The factor 4π in the denominator can be read as the standard two-loop suppression $1/(16\pi^2)$:

$$\varepsilon = \left(\frac{\alpha_G}{4\pi}\right)^2 = \frac{\alpha_G^2}{16\pi^2}. \tag{4.2}$$

This provides a physical interpretation of the ansatz, but does not change its numerical value. The status of ε therefore remains that of an ansatz.

5 Conditionally derived quantities

$$m_{KK} = \frac{M_{Pl}^{\rm red}}{\sqrt{\mathcal{I}}\, e^{\pi^2}} = 2.0838 \times 10^{14}\ \text{GeV}, \tag{5.1}$$

$$M_6 = \left(3(M_{Pl}^{\rm red})^2 m_{KK}^2\right)^{1/4} = 2.9646 \times 10^{16}\ \text{GeV}, \tag{5.2}$$

at $\mathcal{I} = 0.3653$.

6 Axion-like (ALP) sector

6.1 ALP parameters

$$f_a = M_{Pl}^{\rm red}\, \varepsilon^2 = 1.9801 \times 10^{11}\ \text{GeV}, \tag{6.1}$$

$$m_a = \frac{(\chi^{1/4})^2}{f_a} = 28.407\ \mu\text{eV}, \tag{6.2}$$

$$\nu_a = \frac{m_a c^2}{h} = 6.8688\ \text{GHz}. \tag{6.3}$$

Here $\chi^{1/4} \approx 75.0$ MeV is the fourth root of the topological susceptibility of QCD. We emphasize that no explicit $U(1)_{PQ}$ symmetry is introduced; the object is an ALP.

Consequence, not prediction. The numerical values of f_a and m_a follow from the chain $\alpha_G \to \varepsilon \to f_a \to m_a$ (with $\chi^{1/4}$ taken from QCD), and not from a derivation of these quantities from the orbifold geometry.

6.2 ALP-photon coupling

$$g_{a\gamma\gamma} = \frac{\alpha_{\rm EM}}{2\pi f_a}\left(\frac{E}{N} - 1.92\right). \tag{6.4}$$

At $f_a = 1.9801 \times 10^{11}$ GeV: $\alpha_{\rm EM}/(2\pi f_a) = 5.866 \times 10^{-15}\ \text{GeV}^{-1}$.

Table 1: Values of $g_{a\gamma\gamma}$ for the two benchmark scenarios.

Scenario	E/N	$g_{a\gamma\gamma}$, GeV^{-1}
KSVZ	0	-1.126×10^{-14}
DFSZ	8/3	$+4.379 \times 10^{-15}$

6.3 ALP dark matter

$$\Omega_a h^2 \approx 0.12\, \theta_i^2\, f_{\rm anh}(\theta_i) \left(\frac{f_a}{3 \times 10^{11}}\right)^{7/6}, \tag{6.5}$$

where $f_{\rm anh}(\theta_i) = [\ln(e/(1-(\theta_i/\pi)^2))]^{7/6}$ is the anharmonicity factor. Numerical T-effects introduce a +12.4% correction, leading to agreement with Planck at $\theta_i = 1.1162$.

6.4 Black hole superradiance

The ultra-light ALP mass obtained from the chain above has important astrophysical consequences. In the presence of a rotating black hole, the ALP field can extract rotational energy via the superradiance mechanism. The critical black hole mass for which this process is most efficient is given by:

$$M_{\rm BH} \simeq \frac{\hbar c}{G m_a}. \tag{6.6}$$

For the resulting mass $m_a = 28.407\ \mu$eV, this yields:

$$M_{\rm BH} \approx 9.36 \times 10^{24}\ {\rm kg} \approx 4.7 \times 10^{-6}\ M_\odot. \tag{6.7}$$

The superradiance window is an algebraic consequence of the ALP mass m_a; it does not constitute new physics beyond the ALP sector. We note that black holes in this mass range correspond to the primordial black hole (PBH) window. Currently, no direct detections of PBHs with $M \sim 10^{-6} M_\odot$ have been confirmed by LIGO-Virgo-KAGRA. Microlensing surveys (such as OGLE) place strict upper bounds on the fraction of dark matter that can be composed of PBHs in this mass range, but these constraints do not exclude a subdominant population sufficient to test the superradiance effect.

7 Quark sector (11 nominal parameters)

7.1 Matrix structure

$$M_u = \begin{pmatrix} a_u Y_1^{(2)} & b_u Y_3^{(4)} & c_u Y_2^{(6)} \\ a_u Y_3^{(2)} & b_u Y_2^{(4)} & c_u Y_1^{(6)} \\ a_u Y_2^{(2)} & b_u Y_1^{(4)} & c_u Y_3^{(6)} \end{pmatrix},\ M_d = \begin{pmatrix} a_1 Y_1^{(6)} & b Y_3^{(2)} & c Y_2^{(2)} \\ a_1 Y_3^{(6)} & b Y_2^{(2)} & c Y_1^{(2)} \\ a_2 Y_2^{(2)} & b Y_1^{(2)} & c Y_3^{(4)} \end{pmatrix}. \tag{7.1}$$

7.2 Target values

Table 2: Target values (PDG 2024).

Observable	$O_i^{\rm target} \pm \sigma_i$
m_u/m_t	$(1.27 \pm 0.30) \times 10^{-3}/172.0$
m_c/m_t	$(0.620 \pm 0.020)/172.0$
m_d/m_b	$(2.70 \pm 0.20) \times 10^{-3}/2.860$
m_s/m_b	$(55.0 \pm 5.0) \times 10^{-3}/2.860$
m_b/m_t	$(2.860 \pm 0.030)/172.0$
$\lvert V_{us}\rvert$	0.22501 ± 0.00068
$\lvert V_{cb}\rvert$	0.04053 ± 0.00042
$\lvert V_{ub}\rvert$	0.00382 ± 0.00020
$\lvert J\rvert$	$(3.080 \pm 0.130) \times 10^{-5}$

7.3 Results

Independent optimization yields $\chi^2 < 10^{-15}$ at the moduli given in the abstract. The residual χ^2 is the machine-precision limit of the numerical fit, not a physical measurement. The coefficients are:

$$c^{\rm fit} \approx (5.31 \times 10^{-5},\ 7.19,\ 0.039,\ 2.43 \times 10^{-4},\ 0.503,\ 1.37 \times 10^{-3},\ 2.22 \times 10^{-3}). \tag{7.2}$$

Calibration of the overall scale. The overall scale of each mass matrix is unphysical: the observables depend only on mass ratios and CKM elements, which are invariant under $M_u \to kM_u$, $M_d \to kM_d$. We fix $a_u = 1$ without loss of generality.

8 Lepton sector (7 nominal parameters)

The charged lepton masses are taken diagonal. The neutrino mass matrix is constructed from the 3_S contractions of $Y^{(2)}$ and $Y^{(4)}$, while a uniform diagonal shift built from the singlet contractions $S_k = (Y_1^{(k)})^2 + 2Y_2^{(k)}Y_3^{(k)}$ adjusts the absolute neutrino masses.

8.1 Matrix structure

$$M_\nu = g_1 M_{3_S}(Y^{(2)}) + g_2 M_{3_S}(Y^{(4)}) + (g_3 S_2 + g_4 S_4 + g_5 S_6)\,\mathbb{I}, \tag{8.1}$$

with

$$M_{3_S}(Y) = \begin{pmatrix} 2Y_1 & -Y_3 & -Y_2 \\ -Y_3 & 2Y_2 & -Y_1 \\ -Y_2 & -Y_1 & 2Y_3 \end{pmatrix}. \tag{8.2}$$

The uniform diagonal term $(g_3 S_2 + g_4 S_4 + g_5 S_6)\,\mathbb{I}$ shifts the three neutrino masses by a common amount but does not affect the PMNS mixing, which is invariant under $M_\nu \to M_\nu + c\,\mathbb{I}$. The coefficients g_3, g_4, g_5 therefore parametrize absolute neutrino masses, which are not constrained by the oscillation data used here. For the PMNS observables only g_1 and g_2 matter.

8.2 Results

Independent optimization yields the three PMNS angles to high precision:

$$\boxed{\theta_{12} = 33.41^\circ, \quad \theta_{13} = 8.54^\circ, \quad \theta_{23} = 49.10^\circ} \tag{8.3}$$

at

$$\tau_l = -0.8534397135 + 1.6846113968\,i, \quad g_1 = +92.53, \quad g_2 = -164.55, \tag{8.4}$$

with $\chi^2_{\text{angles}} < 10^{-101}$.

Structural prediction $J_{cp} = 0$. The residual CP symmetry of the 3_S contraction combined with the uniform diagonal shift makes the PMNS matrix real up to unphysical field rephasings, so the Jarlskog invariant vanishes identically:

$$\boxed{J_{cp} = 0} \quad \text{(prediction)}. \tag{8.5}$$

We verified this with two independent global optimizations: six seeds across two matrix forms, all converging to $J_{cp} = 0$ while fitting the three angles to high precision. The prediction is not a consequence of a parameter choice; it is a structural property of the model.[2]

[1] The values of g_1 and g_2 displayed in the text are rounded for readability. The exact values used in the computation are $g_1 = +92.5325455271$ and $g_2 = -164.5453654137$. The quoted $\chi^2_{\text{angles}} < 10^{-10}$ refers to these exact values; substituting the rounded coefficients gives angles accurate to $\sim 10^{-5}$.

[2] A formal algebraic proof that the $2/-1$ form admits a residual CP symmetry is not given here. The claim is supported by numerical evidence (six independent DE seeds across two matrix forms, all converging to $J_{cp} = 0$); a complete proof is left to future work.

The current observational central value $J_{cp} \approx -0.018 \pm 0.005$ (NuFIT 5.2) lies at $\approx 3.6\sigma$ from $J_{cp} = 0$. This is a genuine, testable prediction: Hyper-Kamiokande and DUNE will measure J_{cp} to ± 0.01 within the next decade. If $J_{cp} \neq 0$ is established at high significance, this lepton sector is excluded.

Calibration of the overall scale. The overall scale of M_ν is unphysical: the PMNS observables are invariant under $M_\nu \to kM_\nu$. We fix $g_1 = 1$ without loss of generality.

Table 3: PMNS observables (NuFIT 5.2).

Observable	Model	NuFIT 5.2	Status
θ_{12} (solar)	$33.41°$	$33.41° \pm 0.5°$	fitted
θ_{13} (reactor)	$8.54°$	$8.54° \pm 0.1°$	fitted
θ_{23} (atmos.)	$49.10°$	$49.1° \pm 0.5°$	fitted
J_{cp} (CP phase)	0	-0.018 ± 0.005	predicted

9 Full balance

Table 4: Free parameters. Nominal count is 18; two are fixed by overall scale (a_u and g_1), three (g_3, g_4, g_5) parametrize absolute neutrino masses and are not fitted to the observables used here. The remaining 14 parameters (including the ALP misalignment angle θ_i) fit 13 observables and predict one additional observable ($J_{cp} = 0$).

Category	Parameters	Count
Quark moduli	τ_u, τ_d (complex)	4
Quark coefficients	$a_u, b_u, c_u, a_1, a_2, b, c$	7
fixed by calibration	$a_u \equiv 1$	-1
Lepton modulus	τ_l (complex)	2
Lepton coefficients (PMNS)	g_1, g_2	2
fixed by calibration	$g_1 \equiv 1$	-1
Lepton coefficients (masses)	g_3, g_4, g_5	(3, not fitted)
ALP	θ_i	1
Fitted to observables		**14**

Parameter classes. The 14 fitted parameters group into three ontological classes: (i) the six real geometric moduli of the three tori (τ_u, τ_d, τ_l), which are the core of the model; (ii) the seven Lagrangian Yukawa coefficients ($b_u, c_u, a_1, a_2, b, c, g_2$), which are phenomenological at this stage — deriving them from holomorphic 6D $\to$ 4D couplings is the Critical item of the roadmap (§11); (iii) the ALP misalignment angle θ_i, which is a cosmological initial condition rather than a Lagrangian parameter. The three spectator coefficients g_3, g_4, g_5 are not fitted.

$$N_{\text{dof}}^{\text{fit}} = 13 - 14 = -1, \qquad N_{\text{dof}} = (13 + 1) - 14 = 0 \quad (\text{with } J_{cp} = 0 \text{ prediction}). \tag{9.1}$$

Information content. The 13 fitted observables leave $N_{\text{dof}}^{\text{fit}} = -1$. The model additionally predicts the value of a 14th observable, $J_{cp} = 0$ (§8). Counting this prediction

as a falsifiable constraint yields $N_{\text{dof}} = (13+1) - 14 = 0$: the model is overdetermined by one constraint, not underdetermined. We report both numbers; §9.1 uses the fitted-only count $N_{\text{dof}}^{\text{fit}}$, which is the standard input for AIC/BIC.

9.1 Information criteria

Table 5: AIC/BIC for the fitted observables ($n = 13$).

Model	k	χ^2	AIC	BIC
MEML 75	14	$< 10^{-15}$	28.0	35.9
SM	19	0.9	38.9	49.6

We caution that AIC and BIC are derived in the asymptotic regime $n \to \infty$ with fixed k. Here $k/n = 14/13 > 1$, beyond the asymptotic regime; the values are reported for completeness only. If the structural prediction $J_{cp} = 0$ is counted as an additional datum ($n = 14$), then $k/n = 1$, exactly at the boundary — the criteria remain inapplicable in either case. The residuals $< 10^{-15}$ reflect the exactly solvable structure of the fit, not an exceptionally good agreement beyond the model's degrees of freedom.

10 Open problems

1. **Structural prediction $J_{cp} = 0$.** The model predicts $J_{cp} = 0$, which lies at $\approx 3.6\sigma$ from the current NuFIT 5.2 central value $J_{cp} \approx -0.018 \pm 0.005$. This is a testable prediction (Hyper-K, DUNE), not a fit deficiency: no allowed choice of τ_l, g_1, g_2 can produce $J_{cp} \neq 0$.

2. **Coefficient degeneracy.** The coefficients c_i are non-unique: χ^2 is invariant under $c_i \to k c_i$ (Appendix D). This degeneracy is removed by the calibration $a_u = g_1 = 1$ (§7, §8).

3. **Two independent sources of non-identifiability.** The coefficient degeneracy is removed by calibration. The flatness of $\operatorname{Re}\tau$ (Appendix E) is *not* removed and remains an open problem.

4. **Status of α_G.** Postulate. The topological reading $\alpha_G = \chi(S^2)/(N_{\text{fix}}\pi)$ (§3) is a geometric motivation, not a derivation.

5. **Status of ε.** Ansatz; the two-loop reading $\varepsilon = \alpha_G^2/(16\pi^2)$ (§4) is a physical interpretation, not a derivation.

6. **Flatness of $\operatorname{Re}\tau$.** $\operatorname{Re}\tau$ is soft-flat (Appendix E).

7. **LOO instability.** At $N_{\text{dof}}^{\text{fit}} = -1$, spread of $|V_{ub}|$ from -97% to $+25407\%$.

11 Roadmap

Table 6: Phases of development.

Phase	Task	Priority
I	Mechanism for c_i: holomorphic 6D→4D couplings	Critical
I	Flux stabilization of $\mathrm{Im}\,\tau$ ($\pm 3\%$)	High
II	Reduce parameters in quarks (beyond the $a_u = 1$ calibration)	Medium
III	ADMX, DESI, DUNE, KamLAND-Zen, Hyper-K	External

12 Conclusions

What has been done:

1. A joint quark and lepton fit with 14 effective parameters that fit 13 observables and predict one further observable, $J_{cp} = 0$. The model is overdetermined by one constraint: $N_{\mathrm{dof}} = (13 + 1) - 14 = 0$; the fitted-only count is $N_{\mathrm{dof}}^{\mathrm{fit}} = 13 - 14 = -1$.

2. Quark sector: $\chi^2 < 10^{-15}$ (machine limit).

3. Lepton sector: three PMNS angles reproduced to high precision at $\tau_l = -0.8534397135 + 1.6846113968\,i$. Structural prediction $J_{cp} = 0$, currently at $\approx 3.6\sigma$ from the NuFIT 5.2 central value.

4. The ALP mass (from the chain $\alpha_G \to \varepsilon \to f_a$) implies a black hole superradiance window at $M_{\mathrm{BH}} \sim 10^{25}$ kg.

Verdict. The model is phenomenological. It reproduces the 13 fitted observables with high precision, and makes one additional structural prediction ($J_{cp} = 0$) that is testable by next-generation neutrino experiments.

Data Availability

The datasets and the Python verification script generated during and/or analysed during the current study are available in the Zenodo repository at https://doi.org/10.5281/zenodo.23567063 and in the GitHub repository at https://github.com/kolbasa1972/MEML-75-Complete. The script reproduces the numerical results of Sections 3–9.1. The flatness scan of Appendix E was obtained with a separate refit procedure that is not part of the fast verification script.

Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used AI-assisted tools to verify the Python code, refine the theoretical text, and format the LaTeX document. The author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

A A_4 modular forms: full q-expansions

Definition: $q = e^{2\pi i \tau}$.

$$Y_1^{(2)} = 1 + \sum_{n=1}^{17} C_1^{(n)} q^n, \tag{A.1}$$

$$Y_2^{(2)} = -6\, q^{1/3} \left(1 + \sum_{n=1}^{11} C_2^{(n)} q^n\right), \tag{A.2}$$

$$Y_3^{(2)} = -18\, q^{2/3} \left(1 + \sum_{n=1}^{10} C_3^{(n)} q^n\right), \tag{A.3}$$

$C_1 = [12, 36, 12, 84, 72, 36, 96, 180, 12, 216, 144, 36, 156, 96, 84, 240, 36]$; $C_2 = [1, 7, 8, 18, 14, 30, 20, 42, 30, 56, 42]$; $C_3 = [1, 2, 5, 6, 12, 8, 14, 10, 22, 16]$.

Convolutions:

$$Y_1^{(4)} = (Y_1^{(2)})^2 - Y_2^{(2)} Y_3^{(2)}, \tag{A.4}$$

$$Y_2^{(4)} = (Y_3^{(2)})^2 - Y_1^{(2)} Y_2^{(2)}, \tag{A.5}$$

$$Y_3^{(4)} = (Y_2^{(2)})^2 - Y_1^{(2)} Y_3^{(2)}, \tag{A.6}$$

$$Y_1^{(6)} = \tfrac{1}{3}\left[2Y_1^{(2)} Y_1^{(4)} - Y_2^{(2)} Y_3^{(4)} - Y_3^{(2)} Y_2^{(4)}\right], \tag{A.7}$$

$$Y_2^{(6)} = \tfrac{1}{3}\left[2Y_3^{(2)} Y_3^{(4)} - Y_1^{(2)} Y_2^{(4)} - Y_2^{(2)} Y_1^{(4)}\right], \tag{A.8}$$

$$Y_3^{(6)} = \tfrac{1}{3}\left[2Y_2^{(2)} Y_2^{(4)} - Y_3^{(2)} Y_1^{(4)} - Y_1^{(2)} Y_3^{(4)}\right]. \tag{A.9}$$

B Numerical fit procedure

The global fit was performed using `differential_evolution` (SciPy): 4 seeds (42, 43, 44, 45); maxiter= 800; popsize= 25; tol= 10^{-8}.

Quark sector. Bounds: $\mathrm{Re}\,\tau_{u,d} \in [-0.5, 0.5]$, $\mathrm{Im}\,\tau_u \in [0.87, 2.5]$, $\mathrm{Im}\,\tau_d \in [0.87, 1.60]$, $c_i \in [10^{-8}, 10^2]$.

Lepton sector. Bounds: $\mathrm{Re}\,\tau_l \in [-1.5, 1.5]$, $\mathrm{Im}\,\tau_l \in [0.5, 2.5]$, $g_i \in [-300, 300]$.

C Reproduction of τ and χ^2

Table 7: Independent DE runs (quark sector).

Source	$\mathrm{Re}\,\tau_u$	$\mathrm{Im}\,\tau_u$	$\mathrm{Re}\,\tau_d$	$\mathrm{Im}\,\tau_d$
Run 1	−0.15779	2.38518	+0.36939	1.54105
Run 2	−0.15779	2.38518	+0.36939	1.54105
Run 3	−0.12893	2.38411	−0.37212	1.54102

The fitted moduli $\mathrm{Im}\,\tau$ are reproducible across seeds to 10^{-3}. The values of $\mathrm{Re}\,\tau$ are not unique: different seeds land at different points along the soft-flat direction discussed in Appendix E. The quantity χ^2 at the minimum is a machine zero; we quote it as $\chi^2 < 10^{-15}$.

D Proof of coefficient degeneracy

At fixed τ, multiply all c_i by k:

Table 8: Invariance of χ^2 under $c_i \to kc_i$.

k	χ^2
0.5	$< 10^{-15}$
1.0	$< 10^{-15}$
1.87	$< 10^{-15}$
10.0	$< 10^{-15}$

The residual values differ only at the level of numerical noise; the invariance is exact up to machine precision.

E Flatness of $\mathrm{Re}\,\tau$

Procedure. The flatness scan below was obtained by refitting the remaining parameters at each value of $\mathrm{Re}\,\tau$. The corresponding refit cycle is computationally expensive and is not part of the fast verification script, which tests instead the sharpness of the minimum at fixed coefficients.

Scan in $\mathrm{Re}\,\tau_u$: width of $\chi^2 < 10$: $[-0.40, +0.00]$ (0.40). Scan in $\mathrm{Re}\,\tau_d$: width of $\chi^2 < 10$: $[+0.30, +0.50]$ (0.20).

F LOO test

For each $j \in \{1, \ldots, 9\}$: remove the j-th observable, predict the j-th. $N_{\mathrm{dof}} = +1$. Results: mean $|V_{ub}|$ +16728%, std 11819%.

G Systematic search for α_G

Tested $N = 392$ combinations. **2 coincidences below** 0.1%**:** $|Y_1^{(6)}|/|Y_3^{(4)}| = 0.212125$ (-0.0385%); $|Y_2^{(6)}|/\pi = 0.212411$ ($+0.0965\%$). Poisson: $\langle n \rangle = 0.78$, $n_{\mathrm{obs}} = 2$; $P(n \geq 2) = 0.185$.

References

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